

Terna Engineering College

Computer Engineering Department

Class: TE

Sem.: VI

Course: System Security Lab

PART A

(PART A : TO BE REFFERED BY STUDENTS)

Experiment No.13

A.1 Aim: Explore the GPGwin tool and implement email security

A.2 Prerequisite:

1. Basic Knowledge of email, symmetric and asymmetric encryption and decryption

A.3 Outcome:

After successful completion of this experiment students will be able to

To be able to use open source technologies and explore email security and explore various attacks.

A.4 Theory:

Pretty Good Privacy (PGP) is a data encryption and decryption computer program that provides cryptographic privacy and authentication for data communication. PGP is often used for signing, encrypting, and decrypting texts, e-mails, files, directories, and whole disk partitions and to increase the security of e-mail communications.

PGP encryption uses a serial combination of hashing, data compression, symmetric-key cryptography, and finally public-key cryptography; each step uses one of several supported algorithms. Each public key is bound to a user name and/or an e-mail address. The first version of this system was generally known as a web of trust to contrast with the X.509 system, which uses a hierarchical approach based on certificate authority and which was added to PGP implementations later. Current versions of PGP encryption include both options through an automated key management server.

GNU Privacy Guard (GnuPG or GPG) is a free software replacement for Symantec's PGP cryptographic software suite. GnuPG is a hybrid-encryption software program because it uses a combination of conventional symmetric-key cryptography for speed, and public-key cryptography for ease of secure key exchange, typically by using the recipient's public key to encrypt a session key which is only used once. This mode of operation is part of the OpenPGP standard and has been part of PGP from its first version.

PART B

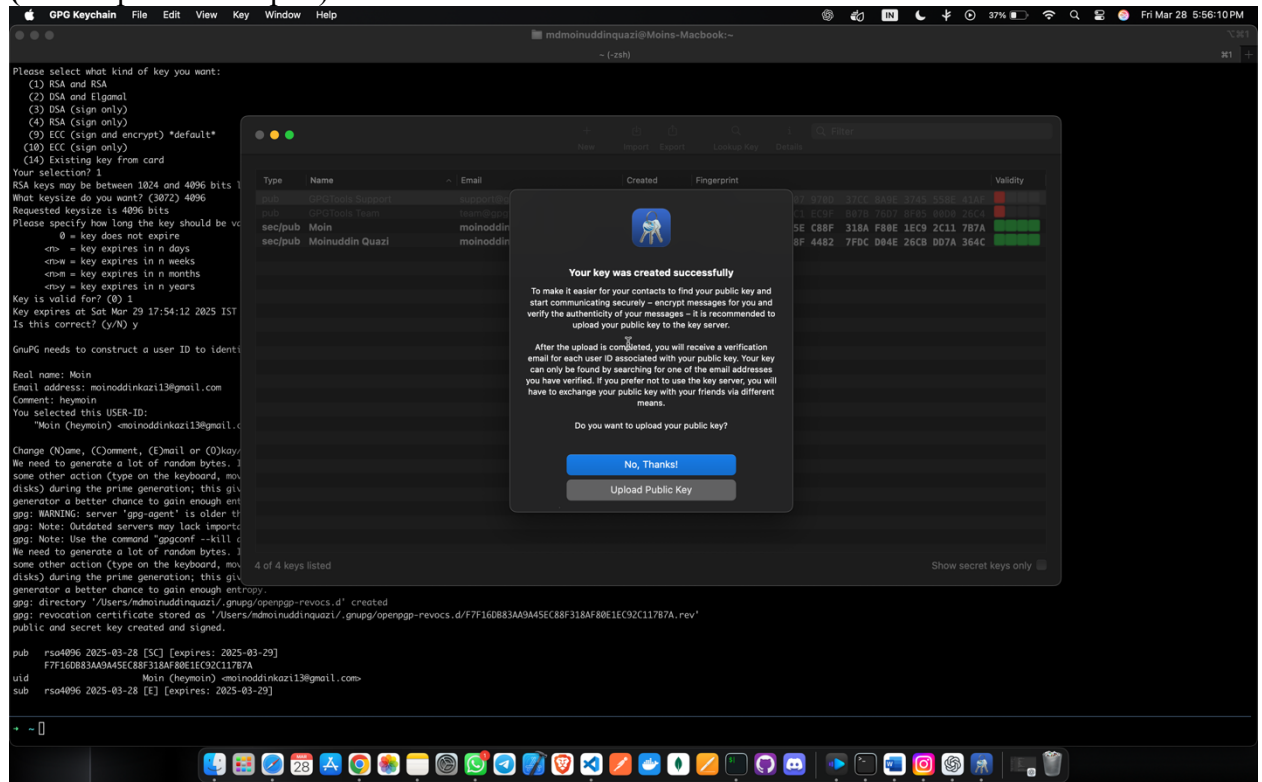
(PART B : TO BE COMPLETED BY STUDENTS)

(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the Blackboard or emailed to the concerned lab in charge faculties at the end of the practical in case there is no Black board access available)

Roll No.C54	Name: Quazi Md Moinuddin
Class:TE-C	Batch:C3
Date of Experiment:	Date of Submission: 28-3-25
Grade :	

B.1 Output

(add snapshot of output)



```
gpg --full-gen-key
Please select what kind of key you want:
(1) RSA and RSA
(2) DSA and Elgamal
(3) DSA (sign only)
(4) RSA (sign only)
(9) ECC (sign and encrypt) *default*
(10) ECC (sign only)
(14) Existing key from card
Your selection? 1
RSA keys may be between 1024 and 4096 bits
What keysize do you want? (3072) 4096
Requested keysize is 4096 bits
Please specify how long the key should be valid
0 = key does not expire
<no> = key expires in n days
<mon> = key expires in n weeks
<mon> = key expires in n months
<mon> = key expires in n years
Key is valid for? (0) 1
Key expires at Sat Mar 29 17:54:12 2025 IST
Is this correct? (y/N) y

GnuPG needs to construct a user ID to identify your key
Real name: Moin
Email address: moinoddinkazi13@gmail.com
Comment: heymain
You selected this USER-ID:
"Moin (heymain) <moinoddinkazi13@gmail.com>"

Change (N)ame, (C)omment, (E)mail or (O)kay/Cancel.
We need to generate a lot of random bytes. It is recommended that you
quit the current program and run something like 'dd if=/dev/urandom of=/dev/null'
some other action (type on the keyboard, move the mouse, etc.) during the
prime generation; this gives the generator a better chance to gain enough
entropy.
gpg: WARNING: server 'gpg-agent' is older than 3.1.2; you should consider
upgrading.
gpg: Note: Outdated servers may lack important functionality.
gpg: Note: Use the command 'gpgconf --kill gpg-agent' to restart the
agent.
We need to generate a lot of random bytes. It is recommended that you
quit the current program and run something like 'dd if=/dev/urandom of=/dev/null'
some other action (type on the keyboard, move the mouse, etc.) during the
prime generation; this gives the generator a better chance to gain enough
entropy.
gpg: directory '/Users/mdmoinuddinquazi/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/Users/mdmoinuddinquazi/.gnupg/openpgp-revocs.d/F7F160B83AA9A45EC88F318AF80E1E392C11787A.rev'
public and secret key created and signed.

pub  rsa4096 2025-03-28 [SC] [expires: 2025-03-29]
     F7F160B83AA9A45EC88F318AF80E1E392C11787A
uid          Moin (heymain) <moinoddinkazi13@gmail.com>
sub  rsa4096 2025-03-28 [E] [expires: 2025-03-29]
```

```

iTerm2 Shell Edit View Session Scripts Profiles Toolbelt Window Help
mdmoinuddinkuazi@Moins-Macbook:~
~ (-zsh)

Please select what kind of key you want:
(1) RSA and RSA
(2) DSA and Elgamal
(3) DSA (sign only)
(4) RSA (sign only)
(9) ECC (sign and encrypt) *default*
(10) ECC (sign only)
(14) Existing key from card
Your selection? 1
RSA keys may be between 1024 and 4096 bits long.
What keysize do you want? (3072) 4096
Requested keysize is 4096 bits
Please specify how long the key should be valid.
  0 = key does not expire
  <n> = key expires in n days
  <nw> = key expires in n weeks
  <nm> = key expires in n months
  <ny> = key expires in n years
Key is valid for? (0) 1
Key expires at Sat Mar 29 17:54:12 2025 IST
Is this correct? (y/N) y

GnuPG needs to construct a user ID to identify your key.

Real name: Moïn
Email address: moïnoddinkazi13@gmail.com
Comment: heymoïn
You selected this USER-ID:
  "Moïn (heymoïn) <moïnoddinkazi13@gmail.com>"

Change (N)ame, (C)omment, (E)mail or (O)kay/(Q)uit? 0
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: WARNING: server 'gpg-agent' is older than us (2.2.41 < 2.4.7)
gpg: Note: Outdated servers may lack important security fixes.
gpg: Note: Use the command 'gpgconf --kill all' to restart them.
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: directory '/Users/mdmoinuddinkuazi/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/Users/mdmoinuddinkuazi/.gnupg/openpgp-revocs.d/F7F16DB83AA9A45EC88F318AF80E1EC92C117B7A.rev'
public and secret key created and signed.

pub  rsa4096 2025-03-28 [SC] [expires: 2025-03-29]
      F7F16DB83AA9A45EC88F318AF80E1EC92C117B7A
uid          Moïn (heymoïn) <moïnoddinkazi13@gmail.com>
sub  rsa4096 2025-03-28 [E] [expires: 2025-03-29]

GPG Keychain File Edit View Key Window Help
mdmoinuddinkuazi@Moins-Macbook:~
~ (-zsh)

(10) ECC (sign only)
(14) Existing key from card
Your selection? 1
RSA keys may be between 1024 and 4096 bits long.
What keysize do you want? (3072) 4096
Requested keysize is 4096 bits
Please specify how long the key should be valid.
  0 = key does not expire
  <n> = key expires in n days
  <nw> = key expires in n weeks
  <nm> = key expires in n months
  <ny> = key expires in n years
Key is valid for? (0) 1
Key expires at Sat Mar 29 17:54:12 2025 IST
Is this correct? (y/N) y

GnuPG needs to construct a user ID to identify your key.

Real name: Moïn
Email address: moïnoddinkazi13@gmail.com
Comment: heymoïn
You selected this USER-ID:
  "Moïn (heymoïn) <moïnoddinkazi13@gmail.com>"

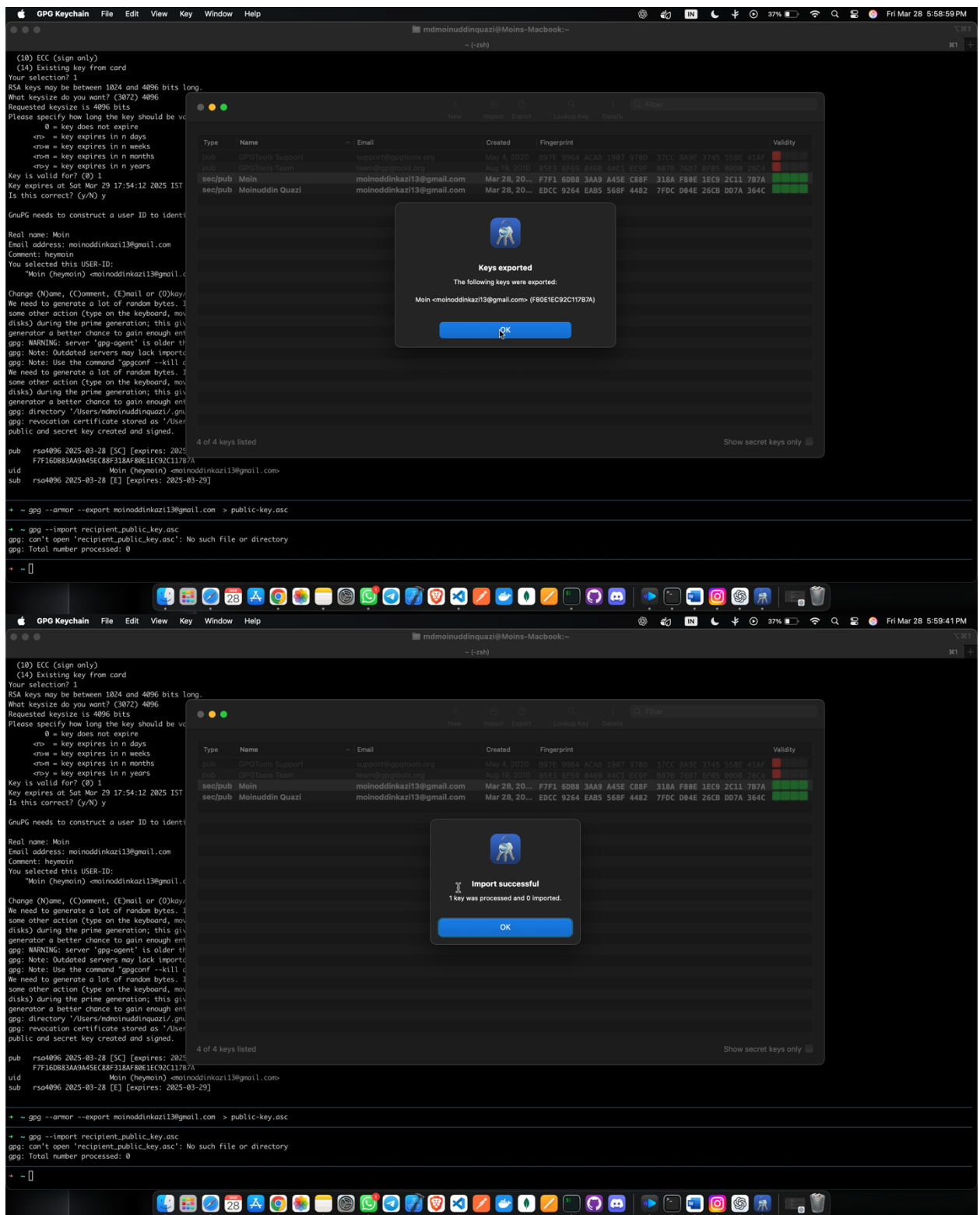
Change (N)ame, (C)omment, (E)mail or (O)kay/(Q)uit? 0
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: WARNING: server 'gpg-agent' is older than us (2.2.41 < 2.4.7)
gpg: Note: Outdated servers may lack important security fixes.
gpg: Note: Use the command 'gpgconf --kill all' to restart them.
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: directory '/Users/mdmoinuddinkuazi/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/Users/mdmoinuddinkuazi/.gnupg/openpgp-revocs.d/F7F16DB83AA9A45EC88F318AF80E1EC92C117B7A.rev'
public and secret key created and signed.

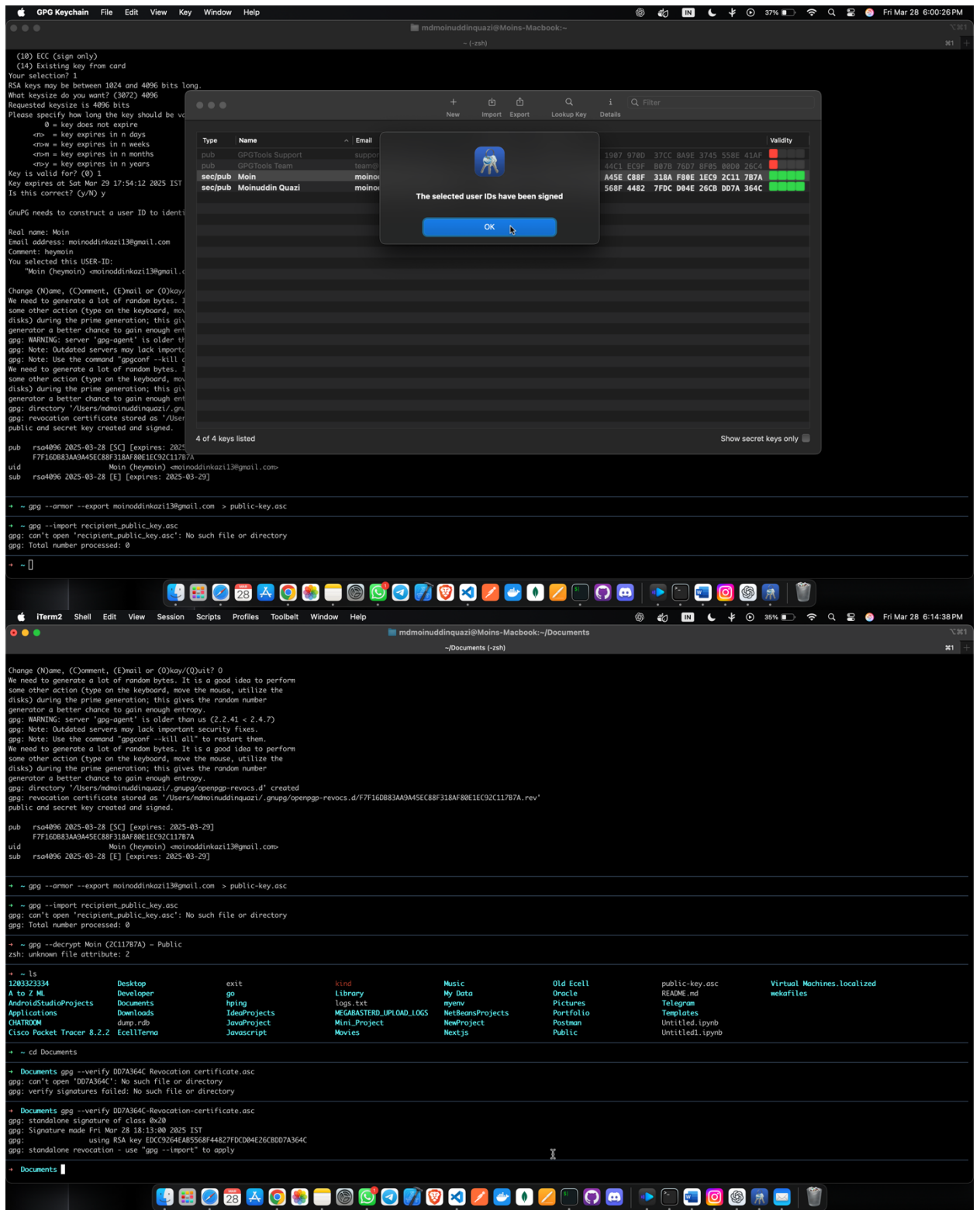
4 of 4 keys listed

+ ~ gpg --armor --export moïnoddinkazi13@gmail.com > public-key.asc
+ ~ gpg --import recipient_public_key.asc
gpg: can't open 'recipient_public_key.asc': No such file or directory
gpg: Total number processed: 0
+ ~

Public keys sent to key server
The following public keys have been uploaded:
Moïnuddin Quazi <moïnoddinkazi13@gmail.com>
(D04E26C8D07A364C)
OK

Type Name Email Created Fingerprint Validity
pub GPGTools Team support@pgptools.org May 8, 2020 857E 960A ACAD 1987 5786 37CC 8A36 3745 558E 4141
pub GPGTools Team team@pgptools.org Aug 16, 2016 85E3 8F59 846B 44C1 EC5F 8676 7807 8F85 86D8 26C4
sec/pub Moïn moïnoddinkazi13@gmail.com Mar 28, 20... F7F1 6DB8 3AA9 A45E C88F 318A F80E 1EC9 2C11 7B7A
sec/pub Moïnuddin Quazi moïnoddinkazi13@gmail.com Mar 28, 20... EDCC 9264 EAB5 56BF 4482 7FDC D84E 26C8 D07A 364C
Show secret keys only
```



B.2. Commands / tools used with syntax:

```
gpg --full-generate-key  
gpg --armor --export your_email@example.com > public-key.asc  
gpg --import recipient_public_key.asc  
gpg --encrypt --recipient recipient@example.com filename.txt  
gpg --decrypt filename.txt.gpg  
gpg --verify signed_file.asc
```

B.3 Question of Curiosity: *(Attempt at least 3 questions handwritten)*

1. What is the full form of PGP?
2. Why email security is required ?
3. What is GPG (GNU Privacy Guard) and how does it ensure email security?
4. What is the difference between symmetric and asymmetric encryption in the context of GPG?
5. What is the role of a public key and private key in email encryption with GPG?
6. What is the process for verifying the authenticity of an email using GPG?

Q1]

What is the full form of PGP?

→ PGP (Pretty Good Privacy) is an encryption program that provides cryptographic privacy & authentication for emails & other digital communications. It was delivered by Phil Zimmermann in 1991 & is widely used for securing sensitive information.

Q4]

What is the difference between symmetric & asymmetric encryption in the context of GPG?

→ • Symmetric encryption: Uses a single secret key for both encryption & decryption. It is fast & efficient, but sharing the key securely is a challenge.

• Asymmetric encryption: Uses two mathematically related keys - a public key & private key. This method is more secure for email communication as it eliminates the need to share a single key.

Q5]

What is the role of a public key & private key in email encryption with GPG?

→ • Public key: This key is shared with others so they can encrypt emails sent to you. It ensures that only the intended recipient can read the message.

• Private key: This key is kept secret & is used to decrypt emails encrypted with your public key. It is also used for digitally signing emails to prove authenticity & integrity, ensuring the recipient knows the email is genuinely from you.

B.4 Conclusion:

(Write appropriate conclusion.)

You have successfully set up **email encryption** and **file security** using **GPG**. Now, we can securely communicate by encrypting messages and verifying sender authenticity.